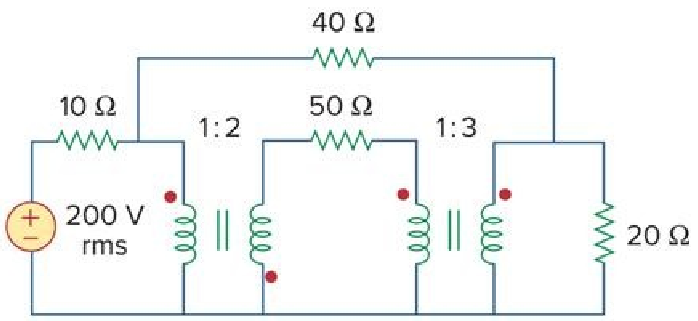
Chapter 13, Problem 65P

Problem

Calculate the average power dissipated by the 20-Ω resistor in Fig. 13.130.

Figure 13.130



Step-by-step solution

Step 1 of 5

Refer to Figure 13.129 in the textbook.

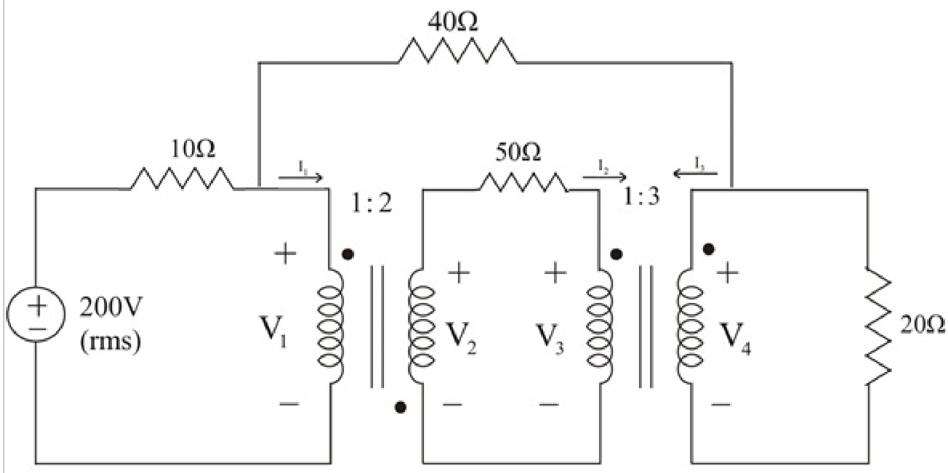


Figure 1

Step 2 of 5

Apply Kirchhoff's current law at node 1.

$$\frac{200-V_1}{10}=\frac{V_1-V_4}{40}+I_1$$
$$4(200-V_1)=V_1-V_4+40I_1$$

$$5V_1-V_4+40I_1=800 \quad \ldots\ldots (1)$$

Apply Kirchhoff's current law at node 2.

$$\frac{V_1-V_4}{40}=\frac{V_4}{20}+I_3$$

$$V_1=3V_4+40I_3 \quad \ldots\ldots (2)$$

Calculate the ratio of primary voltage to secondary voltage of first transformer.

$$\frac{V_2}{V_1}=-\frac{N_2}{N_1}$$
$$=-2$$

$$V_2=-2V_1 \quad \ldots\ldots (3)$$

$$\frac{I_2}{I_1}=-\frac{N_1}{N_2}$$
$$=-\frac{1}{2}$$

$$I_1=-2I_2 \quad \ldots\ldots (4)$$

Apply Kirchhoff's voltage law for the middle loop.

$$-V_2+50I_2+V_3=0$$

$$V_3=V_2-50I_2 \quad \ldots\ldots (5)$$

Step 3 of 5

Calculate the ratio of currents and voltages at the terminals of second transformer.

$$\frac{V_4}{V_3}=\frac{N_2}{N_1}$$
$$=3$$

$$V_4=3V_3 \quad \ldots\ldots (6)$$

$$\frac{I_3}{I_2}=-\frac{N_1}{N_2}$$
$$=-\frac{1}{3}$$

$$I_2=-3I_3 \quad \ldots\ldots (7)$$

Step 4 of 5

From equations (1) and (2),

$$14V_4 + 200I_3 + 40I_1 = 800$$

Substitute $3I_3$ for I_2 in equation (4).

$$\begin{aligned} I_1 &= -2I_2 \\ &= -2(3I_3) \\ &= -6I_3 \end{aligned}$$

$$14V_4 + 200I_3 + 40(-6I_3) = 800$$
$$14V_4 - 40I_3 = 800$$

$$V_3 = V_2 - 50I_2$$

$$\begin{aligned} V_4 &= 3(V_2 - 50I_2) \\ &= 3V_2 - 150I_2 \\ &= 3(-2V_1) - 150(-3I_3) \end{aligned}$$

$$\begin{aligned} V_4 &= -6V_1 + 450I_3 \\ &= -6(3V_4 + 40I_3) + 450I_3 \\ &= -18V_4 + 210I_3 \end{aligned}$$

$$\begin{aligned} 19V_4 &= 210I_3 \\ I_3 &= \frac{19}{210}V_4 \end{aligned}$$

Substitute $\frac{19}{210}V_4$ for I_3 in equation (1).

$$\begin{aligned} 200 &= 13.452V_4 \\ V_4 &= 14.87 \end{aligned}$$

Step 5 of 5

The average power dissipated by the $20\,\Omega$ resistor is,

$$\begin{aligned} P_{20\,\Omega} &= \frac{V_4^2}{20} \\ &= \frac{14.87^2}{20} \\ &= 11.05\text{ W} \end{aligned}$$

The average power dissipated by the $20\,\Omega$ resistor is $\boxed{11.05\text{ W}}$.